

ABIR DAS

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🔗 [LinkedIn](#) · [GitHub](#) · [Resume Website](#)

ABOUT ME

AI/ML Engineer with 3+ years of applied experience in deep learning, computer vision, and medical imaging. First author on 4 research papers (2 published), with expertise in supervised learning, Vision Transformers, self-supervised learning, and model deployment. Proven ability to translate research into real-world solutions across domains like healthcare and speech processing. Proficient in PyTorch, TensorFlow, scikit-learn, Hugging Face, and AWS. Passionate about explainable AI and practical innovation in AI-driven systems.

WORK EXPERIENCE

AI Software Developer Intern
NINANO Company Inc

(Sep 2024 - Nov 2024)

📍 Gimcheon-si, Gyeongsangbuk-do, Republic of Korea.

Autonomous Obstacle Avoidance System Using Stereo Vision

- Developed a stereo vision-based obstacle detection system using MiDaS and SGBM algorithms, enabling accurate depth perception up to 8 meters.
- Integrated the system with a drone's flight controller to trigger real-time safety braking and operator notifications.
- Improved low-altitude obstacle response reliability by 30%, enhancing drone stability at a flight height of 10 meters.
- Technologies:** MiDaS, SGBM, YOLO, Python, PX4 Autopilot, Embedded Control, NVIDIA Jetson Orin Nano

Sub-Project:

Gimbal Camera Object Tracking System

- Enhanced YOLO object detection for a drone-mounted gimbal camera to autonomously track a primary target while detecting multiple secondary objects within the same frame.
- Boosted object tracking precision and real-time responsiveness during live aerial missions.
- Built a target-prioritization module to improve contextual awareness of the visual field.
- Technologies:** YOLO, Python, Computer Vision, Gimbal Control Systems

Embedded Systems Intern
SpaceK

(Dec 2022 - Jan 2023)

📍 Chungnam National University Start-up Support Center, Daejeon, Republic of Korea.

- Designed and implemented embedded software solutions for satellite communication systems, improving data transmission efficiency by 20% and enhancing control reliability.
- Developed microcontroller-based solutions using Embedded Linux, UART, I2C, and SPI protocols to streamline satellite communication processes.

RESEARCH EXPERIENCE

Research Assistant | MSPI Research Group | [01-2023] – [10-2024]

- Conducted advanced research in brain-computer interface (BCI) using motor imagery EEG signals.
- Authored a paper accepted by Scientific Reports, contributing to model design, signal preprocessing, and manuscript writing.
- Explored hybrid deep learning architectures for real-time mental state classification.

Site - <https://sites.google.com/view/drjiauddin/research-areas?authuser=0>

Research Assistant | Chungnam National University | [12-2024] – [Present]

- Led research on computer vision and fault diagnosis using explainable AI (XAI), MiDaS, and YOLO for industrial applications.
 - Published a paper in Computers, Materials & Continua and contributed to XAI-based visual interpretation pipelines.
 - Supported peer-review and technical writing processes.
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PROJECTS

MediScan – Skin Disease Detection

Built a mobile-friendly CNN-based AI system to detect 7+ skin diseases. Achieved 92% accuracy with Grad-CAM explainability. **Tech:** PyTorch, OpenCV, Grad-CAM, Streamlit, Hugging Face

Real-Time Accent Change Generator (Ongoing)

Developing a live speech-to-speech conversion system for accent translation from Indian to US English using Whisper and Tacotron2. **Tech:** PyTorch, Whisper, Tacotron2, WaveGlow, Flask, Streamlit

GitHub Resume Website

Designed and deployed a personal website to showcase research, projects, and contact details. **Tech:** HTML, CSS, GitHub Pages

Car Parking Occupancy Detection

Built a YOLOv3-based system to detect parking slot availability using real-time camera feeds. Achieved 90% accuracy. **Tech:** YOLOv3, OpenCV, Python

Facial Expression Recognition

Developed a CNN to classify 7 emotional states from the FER-2013 dataset. Achieved 92% accuracy. **Tech:** TensorFlow, Python

Boston Housing Data Visualizer

Built a Tableau dashboard using clustering (K-Means) and regression (Linear) models to analyze Boston housing data. **Tech:** Tableau, Python

Scalable ML Model Deployment on AWS

Deployed an ML pipeline on AWS SageMaker, reducing training time by 40% and enabling real-time inference. **Tech:** AWS SageMaker, Python, scikit-learn

JCFS Assistant Chatbot

Built a chatbot using Flask and Google NLP API for academic query handling. Reduced manual query load by 50%. **Tech:** Flask, Google NLP, Sentiment Analysis

Real-Time Object Detection on 52-Card Deck

Created a real-time system for recognizing playing cards using OpenCV and contour analysis with 98% accuracy. **Tech:** Python, OpenCV

EDUCATION

Woosong University, South Korea

Bachelor of Science in Artificial Intelligence - 3.83 - 4.5 / 86.6%

March 2021 - February 2025

Siliguri Government Polytechnic, India

Diploma in Electronics & Instrumentation Engineering - 70%

August 2016 - July 2019

Kanchrapara Harnett High School

High School - 79%

January 2011 - March 2016

SKILLS & EXPERTISE

Programming & Scripting:

Python, Bash, SQL

Deep Learning & ML Frameworks:

PyTorch, TensorFlow, Keras, scikit-learn, Hugging Face Transformers, OpenCV, Grad-CAM, CLIP, CUDA, NumPy, Pandas

ML Techniques & Domains:

Supervised Learning, Transfer Learning, Vision Transformers (ViT), Self-Supervised Learning, Few-Shot & Zero-Shot Learning, Contrastive Learning, Explainable AI (XAI), CNNs, RNNs, Object Detection (YOLO, MiDaS, SGBM), Signal Processing, Sentiment Analysis

MLOps & Deployment:

AWS SageMaker, Flask, Streamlit, GitHub Pages, REST APIs, Docker (basic), Git, CI/CD (basic)

Visualization & Data Tools:

Tableau, SPSS, Jupyter, Google Colab, Anaconda

Hardware & Embedded Systems:

NVIDIA Jetson Orin Nano, Raspberry Pi 5, PX4 Autopilot, Embedded Linux, UART, I2C, SPI

CERTIFICATIONS

- **Introduction to TensorFlow for AI, ML, and DL** - Coursera
 - **Neural Networks and Deep Learning** - Coursera
 - **Machine Learning with Python (IBM)** - Coursera
 - **Digital Accounting Transformation** – Partnership of Asian Management School
 - **Python** – Partnership of Asian Management School
 - **Python Programming** – Microsoft, Kolkata
 - **Android App Development** – IIT Kharagpur
 - **Data Visualization** – Partnership of Asian Management School
 - **UX Design** – Partnership of Asian Management School
 - **Elice Game Development in Python** – Korea
 - **Research Assistant Certificate** – Chungnam National University
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PUBLICATIONS

- **Machine Fault Diagnosis Using Sensor Data and Explainable AI Techniques** – Computers, Materials & Continua (CMC) Journal - <https://www.techscience.com/cmc/v80n3/57907>
 - **Enhanced EEG Signal Classification in Brain Computer Interfaces Using Hybrid Deep Learning Models** – Scientific Reports - <https://www.nature.com/articles/s41598-025-07427-2>
 - **Few-Shot and Zero-Shot Brain Tumour Classification using CLIP & ViT** - Under review at International Journal of Sensors, Wireless Communications and Control
 - **Self-Supervised Multi-Label Vision Transformer for Chest X-Ray Diagnosis** - To be submitted to a journal with IF > 8
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